

## 2.2 Programming fundamentals

Why this unit is taught the way it is

### What this unit teaches, and where it ends

8 lessons: 7 with an online experience, 1 paper assessment and 0 bonus challenges. 99 activities in all, worth 135 marks.

By the end of it a pupil should be able to:

- know what variables and constants are, and how a value is assigned to each [pf-101](#)
- understand how input and output move data in and out of a program [pf-101](#)
- know what sequence, selection and iteration each do to the order a program runs in [pf-101](#)
- know the data types integer, real, Boolean, character and string [pf-102](#)
- understand why casting is needed and how a value is cast from one type to another [pf-102](#)
- know how the arithmetic, comparison and Boolean operators are used [pf-102](#)
- know why a program needs to store data in a file [pf-103](#)
- understand how a file is opened, read from, written to and closed [pf-103](#)
- know the difference between writing to a file and appending to it [pf-103](#)
- know what is meant by a record, a field and a table of records [pf-104](#)
- understand how SELECT, FROM and WHERE are used to search for data [pf-104](#)
- know how AND, OR, NOT and LIKE narrow a search down [pf-104](#)
- know how one- and two-dimensional arrays are declared, read and changed [pf-105](#)
- understand why arrays are zero-indexed and how a loop iterates over one [pf-105](#)
- know the difference between a procedure and a function, and between local and global variables [pf-105](#)
- know what random numbers are used for in a program [pf-106](#)
- understand how a random whole number is generated within a given range [pf-106](#)
- know that a computer produces pseudo-random numbers from an algorithm and a seed [pf-106](#)
- know the whole of topic 2.2 well enough to answer without notes [pf-107](#)
- understand the mistakes that most often lose marks in programming questions [pf-107](#)
- know which lessons are secure and which still need revision before the test [pf-107](#)

### Why the lessons are in this order

Each lesson's starter retrieves the one before it, so the order is the retrieval schedule as much as the teaching one. The chain for this unit:

| LESSON                   | TEACHES                                                                    | RETRIEVES                      |
|--------------------------|----------------------------------------------------------------------------|--------------------------------|
| <b>1</b> Variable store  | know what variables and constants are, and how a value is assigned to each | nothing before it in this unit |
| <b>2</b> Type foundry    | know the data types integer, real, Boolean, character and string           | Variable store                 |
| <b>3</b> Filing room     | know why a program needs to store data in a file                           | Type foundry                   |
| <b>4</b> Record vault    | know what is meant by a record, a field and a table of records             | Filing room                    |
| <b>5</b> Array yard      | know how one- and two-dimensional arrays are declared, read and changed    | Record vault                   |
| <b>6</b> Chance machine  | know what random numbers are used for in a program                         | Array yard                     |
| <b>7</b> Revision HQ 2.2 | know the whole of topic 2.2 well enough to answer without notes            | Chance machine                 |

| LESSON              | TEACHES                 | RETRIEVES       |
|---------------------|-------------------------|-----------------|
| 8 End-of-topic test | assesses the whole unit | Revision HQ 2.2 |

The assessment in this unit is placed where it is so that the reflection sheet still has lessons left to send a pupil back to.

## What pupils find hard here

None of this unit's walls names a common mistake in so many words. The misconceptions it works against are in the wrong options of its questions instead: every one of those is a plausible wrong idea rather than filler, and the lesson plans list them question by question.

Every multiple-choice and Predict activity in the unit — 92 of them — offers wrong answers that are the misconception, not filler. The split across those options at the hinge question is the cheapest diagnostic in the lesson.

## Where the explaining, modelling and practising happen

| PHASE           | WHERE IT IS                                                                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Retrieve</b> | The starter on the worksheet, before any device is opened. It is answered on paper and corrected before the lesson moves on.                                                                                                                        |
| <b>Explain</b>  | The slides and the station walls carry the same explanation, so a teacher can teach from the front and a pupil can re-read it in the room.                                                                                                          |
| <b>Model</b>    | There is no worked example on screen in this unit: its activities are questions to answer rather than techniques to copy. The modelling is yours, from the slides, before anyone starts - and it is the part to protect when the lesson runs short. |
| <b>Practise</b> | 94 guided activities, then 5 independent ones.                                                                                                                                                                                                      |
| <b>Check</b>    | The badge questions mark as the pupil goes; the hinge question is for the class at once.                                                                                                                                                            |
| <b>Apply</b>    | 0 challenge activities, and the exam practice on the worksheet.                                                                                                                                                                                     |
| <b>Review</b>   | The exit question and the confidence columns. The confidence columns supplement the evidence; they do not stand in for it.                                                                                                                          |

## How the website, the slides and the worksheet divide the work

They are not three copies of the same thing.

- **The slides** are for the front of the room: the explanation and the model, before anyone is working alone.
- **The experience** is where a pupil reads, decides and is marked. It gives the immediate feedback a teacher cannot give thirty pupils at once.
- **The worksheet** is the record and the writing. The key fact in their own words, the station challenge, the exam practice. It is also what survives when the devices do not.

The rule that holds them together is **write first, then answer at the badge**. A pupil who taps first has turned the lesson into a quiz game, and the worksheet is how you can see at a glance who did.

## Why a 360° room, and what it should produce

The room is not there to be impressive. It is there because this unit's content is a set of related parts that a pupil has to hold together, and putting one part on each wall makes the set visible and navigable: a pupil can turn back to station 2 while answering station 4. The evidence it should produce is on paper — 42 key facts and 42 station challenges in a pupil's own words — and in the score screen, which shows which station a pupil lost marks on rather than one number.

No claim is made here that immersion improves learning on its own. What the format does is make the structure of a topic physical and keep a pupil's attention on one part at a time; the teaching is in the explanation, the example and the practice, as it would be anywhere else.

## Support, and how it is taken away

- The worked example is on the early activities of a station and not on the late ones. That fade is deliberate.

- The hint is a ladder asked for one rung at a time, and the top of it is still the technique on different data, never the answer.
- Read aloud is on every task.
- Prescribed values are boxed and in a monospace face as well as coloured, so they survive a black-and-white printout and a high-contrast mode.
- **Lower literacy and EAL:** pre-teach the vocabulary (Assignment, Count-controlled iteration, Casting, Concatenation, File handling, Append, Record, Wildcard, Zero-indexing, Sub-program, Pseudo-random, Seed). Accept the key fact spoken before it is written, and in two or three words.
- **SEND:** pace is not fixed. Progress saves after every answer and reopens at the earliest unfinished question, so a pupil may finish a station next lesson without losing anything.
- A headset is never required.

## Checking, and what to do with what you find

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Three checks, in this order: the starter (did last lesson stick), the hinge question (is the class ready for independent work), the exit question (did this lesson stick). The first two change what you do in the lesson; the third changes what you start the next one with.

When a hinge question splits the class, go back to the station it sits on and work its example again, aloud, before anyone goes on. When a test comes back, the reflection sheet names the topics to revisit and review mode on those lessons reopens exactly the questions a pupil got wrong.

## Where the approach comes from

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The shape of a lesson here — retrieve, explain, model, practise with support that fades, check, apply, review — is ordinary evidence-informed practice rather than anything this product invented. Useful starting points, none of them about Revise 360:

- Rosenshine, *Principles of Instruction* (American Educator, 2012) — the daily review, small steps, worked examples and checking for understanding that this structure follows.
- Sweller, van Merriënboer and Paas on cognitive load and the worked-example effect — why the example comes before the independent task and why the support is withdrawn gradually.
- Dunlosky et al., *Improving Students' Learning With Effective Learning Techniques* (2013) — the evidence for retrieval practice and spacing, which the starters and the review mode use.
- Wiliam, *Embedded Formative Assessment* — hinge questions and what to do with the answers.

**What is not claimed.** Nothing above is evidence that Revise 360 improves outcomes. No trial of this product has been run, and immersion is not claimed to improve learning by itself. The citations are for the teaching structure; whether these resources work in your classroom is a question your own assessment answers.

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